

On-line Rigid Object Recognition and Pose Estimation Based on Inertial Parameters

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Abstract—This paper proposes an object recognition and gripping pose estimation approach based on on-line estimation of the complete set of inertial parameters, i.e. the mass, the coordinates of the center of mass, and the elements of the inertia matrix, of an object gripped by or attached to a manipulator. A multi-sensor fusion approach combining 6D force/torque, 6D acceleration, 3D angular velocity, and joint angle data to estimate these parameters is presented. In order to facilitate practical implementation, approaches to handling force/torque sensor offsets and to compensating the forces/torques caused by the distal mounting plate of the force/torque sensor and the gripper are incorporated. Regarding the joint angle signals, preprocessing steps to derive the angular velocity, linear acceleration and angular acceleration vector w.r.t. the sensor frame are addressed. The estimation of the complete set of inertial parameters employing the recursive instrumental variables (RIV) method is discussed. The extraction of features that are invariant w.r.t. translation and rotation, i.e. the mass and the principal moments of inertia, as well as a recognition approach based on the Kullback-Leibler divergence are presented. Experimental results show very low errors in the estimates of the inertial parameters, good pose estimation accuracy, and the viability of the recognition approach.

I. INTRODUCTION

Most 3D rigid object recognition approaches rely on vision information. Regarding model-based object recognition and classification, sensor data are used to extract features that may be related to the shape, the color, and/or the texture of the object. Apart from these classes of *visual* features, however, inertial parameters of an object may serve as features as well. In robotics applications it may be desirable to determine which object of a set of objects has been gripped by a manipulator or to check whether the inertial parameters of a gripped object are within given tolerances. Purely vision based approaches may fail if the objects to be inspected are similar with respect to their visual characteristics. Recognition or classification methods based on inertial features (or combinations of visual and inertial features) can overcome this shortcoming. For example, consider a casting with blowholes. Visual inspection of the casting cannot detect the blowholes whereas its inertial features can reveal the manufacturing error.

A desirable property of a feature, regardless whether it is based on visual or inertial characteristics, is invariance w.r.t. translation and rotation of the examined object. Regarding the complete set of inertial parameters (mass, coordinates of the center of mass, and the inertia matrix) of

an object, four features that are invariant w.r.t. translation and rotation can be extracted: the mass and the principal moments of inertia. In robotics applications, inertial parameters of an object gripped by a manipulator may be estimated by moving it on a special trajectory. If this trajectory is selected appropriately, the estimation of all ten inertial parameters - and hence object recognition utilizing the four features mentioned above - can be performed in a single experiment.

While the estimation of general robot dynamic parameters has received considerable attention over the last decades, the estimation of the complete set of inertial parameters of an object attached to a manipulator has been addressed by few researchers. An et al. [1] addressed the problem of estimating inertial load parameters employing a wrist-mounted force/torque sensor. Kozłowski [2] extended this approach by the derivation of optimized excitation trajectories. The latter approaches employed industrial manipulators whereas Niebergall [3] developed a 3-DoF measurement robot solely dedicated to the estimation of the complete set of inertial parameters of an object.

The herein proposed approach relies on a wrist-mounted force/torque sensor as well but incorporates additional sensor data. In contrast to the estimation approaches mentioned above, it fuses angular velocity and acceleration signals from different sensors to effectively reduce estimation bias. Inertial parameters estimation, pose estimation, and object recognition are performed on-line during trajectory execution. Moreover, the handling of sensor offsets and the compensation of forces/torques caused by the mounting plate of the force/torque sensor and the gripper are integrated in the estimation approach. Having estimated the complete set of inertial parameters of the object employing the recursive instrumental variables (RIV) method, a four dimensional feature vector containing the object mass and the principal moments of inertia, and its covariance matrix are computed utilizing the scaled unscented transform [4]. The herein proposed recognition process relies on the popular Kullback-Leibler divergence [5] as distance measure. Based on the eigenvectors of the inertia matrices in different object poses, relative transformations between these poses can be computed. These transformations may be used to correct gripping errors in assembly tasks prior to mating.

The rest of the paper is organized as follows. In Section II the equations employed in the parameter estimation step are

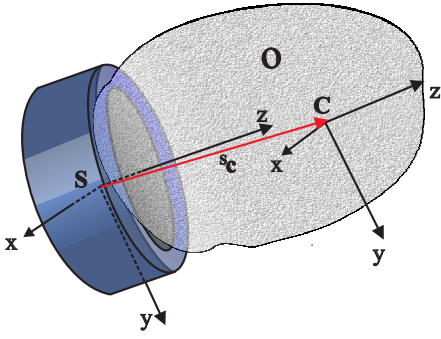


Fig. 1. Force/torque sensor with attached object.

addressed. Section III briefly outlines the design approach for the employed excitation trajectories. In Section IV the pre-processing stage is described. Section V covers the selected parameter estimation approach. The extraction of features from the complete set of inertial parameters is described in Section VI. The recognition approach and the estimation of gripping poses are addressed in Section VII. Section VIII presents various experimental results of the trajectory design, the inertial parameters estimation, the recognition, and the pose estimation approach.

II. EQUATIONS FOR PARAMETER ESTIMATION

The equations used to estimate the inertial parameters (mass m , coordinates of the center of mass c , and inertia matrix I) of the object are derived from the basic laws of dynamics. Based on the Newton-Euler approach, the motion of a rigid body due to external forces and torques is described by two vector equations, i.e. eqns. 1 and 2, that are *linear* w.r.t. to the unknown parameters. Fig. 1 depicts the force/torque sensor with an object rigidly attached to it. The measured forces ${}^S f$, torques ${}^S \tau$, the linear and angular acceleration vector ${}^S a$ and ${}^S \alpha$, the angular velocity vector ${}^S \omega$, the gravity vector ${}^S g$, and the inertia matrix ${}^S I$ refer to the sensor frame S . ${}^S a$ denotes the linear acceleration of the sensor frame due to translation. Note that the axes of the frame C located at the center of mass (COM) of the object are parallel to the axes of S .

$${}^S f = m {}^S a - m {}^S g + {}^S \alpha \times m {}^S c + {}^S \omega \times ({}^S \omega \times m {}^S c) \quad (1)$$

$${}^S \tau = {}^S I {}^S \alpha + {}^S \omega \times ({}^S I {}^S \omega) + m {}^S c \times {}^S a - m {}^S c \times {}^S g \quad (2)$$

where

$$I = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{pmatrix} \quad (3)$$

The matrix equation 4 used for parameter estimation relates the measured forces/torques ${}^S f$, ${}^S \tau$ and the variables ${}^S a$, ${}^S \alpha$, ${}^S \omega$, and ${}^S g$ to the complete set of inertial parameters contained in ${}^S \varphi$.

$$\begin{pmatrix} {}^S f \\ {}^S \tau \end{pmatrix} = V {}^S \varphi \quad (4)$$

Equations 1 and 2 are used to compose the desired matrix V , eq. 5 [2]. Note that the index S is omitted in eq. 5. The

parameter vector ${}^S \varphi$ contains the complete set of inertial parameters of the object:

$${}^S \varphi = [m, m {}^S c_x, m {}^S c_y, m {}^S c_z, {}^S I_{xx}, {}^S I_{xy}, {}^S I_{xz}, {}^S I_{yy}, {}^S I_{yz}, {}^S I_{zz}]^T \quad (6)$$

The matrix V , as defined above, does not consider an important characteristic of common strain gage force/torque sensors. The forces and torques measured by these sensors typically show offsets that would deteriorate the estimation results. In order to handle these offsets two approaches are pursued:

- Zeroing the sensor values in a known sensor orientation and considering the eliminated gravitational forces and associated torques in the estimation process by a pseudo gravity vector ${}^S g_{init} = [{}^S g_{init_x}, {}^S g_{init_y}, {}^S g_{init_z}]^T$.
- Directly estimating the sensor offsets.

The first approach requires zeroing each time an object is gripped. The second approach does not entail any zeroing but the number of parameters to be estimated increases from 10 to 16. The first approach leads to the matrix V_{ginit} which is composed of a matrix $\bar{V}_{ginit}$ that considers the gravitational forces and associated torques that are eliminated by zeroing the sensor values and a 6×6 zero matrix.

$$\bar{V}_{ginit} = \begin{pmatrix} {}^S g_{init_x} & 0 & 0 & 0 \\ {}^S g_{init_y} & 0 & 0 & 0 \\ {}^S g_{init_z} & 0 & 0 & 0 \\ 0 & 0 & {}^S g_{init_z} & -{}^S g_{init_y} \\ 0 & -{}^S g_{init_z} & 0 & {}^S g_{init_x} \\ 0 & {}^S g_{init_y} & -{}^S g_{init_x} & 0 \end{pmatrix} \quad (7)$$

$$V_{ginit} = [\bar{V}_{ginit} \mathbf{0}_{6 \times 6}] \quad (8)$$

The resulting matrix V_{offs} used for parameter estimation is the sum of eq. 5 and eq. 8.

$$V_{offs} = V + V_{ginit} \quad (9)$$

To explicitly identify the force/torque offsets, the matrix V is augmented by an identity matrix E and the parameter vector ${}^S \varphi$ is expanded by the force/torque offsets to be estimated.

$${}^S \varphi_{ext} = [f_{ox}, f_{oy}, f_{oz}, \tau_{ox}, \tau_{oy}, \tau_{oz}, {}^S \varphi]^T \quad (10)$$

$$V_{ext} = [E_{6 \times 6} V] \quad (11)$$

Acceleration sensors may exhibit offsets as well. In this case, the values of the linear acceleration sensor are set to the current gravity vector ${}^S g$ derived from joint angle measurements before trajectory execution and the values of the angular acceleration sensor are zeroed.

III. TRAJECTORIES FOR PARAMETER ESTIMATION

To allow for the estimation of *all ten* inertial parameters of an object in a single experiment, the employed trajectory has to meet requirements concerning parameter excitation. For instance, a linear movement of the sensor frame does not permit estimating the elements of the inertia matrix. Regarding the influence of the selected trajectory on the

$$V = \begin{pmatrix} a_x - g_x & -\omega_y^2 - \omega_z^2 & \omega_x \omega_y - \alpha_z & \omega_x \omega_z + \alpha_y & 0 & 0 & 0 & 0 & 0 & 0 \\ a_y - g_y & \omega_x \omega_y + \alpha_z & -\omega_x^2 - \omega_z^2 & \omega_y \omega_z - \alpha_x & 0 & 0 & 0 & 0 & 0 & 0 \\ a_z - g_z & \omega_x \omega_z - \alpha_y & \omega_y \omega_z + \alpha_x & -\omega_y^2 - \omega_x^2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & a_z - g_z & g_y - a_y & \alpha_x & \alpha_y - \omega_x \omega_z & \alpha_z + \omega_x \omega_y & -\omega_y \omega_z & \omega_y^2 - \omega_z^2 & \omega_y \omega_z \\ 0 & g_z - a_z & 0 & a_x - g_x & \omega_x \omega_z & \alpha_x + \omega_y \omega_z & \omega_z^2 - \omega_x^2 & \alpha_y & \alpha_z - \omega_x \omega_y & -\omega_x \omega_z \\ 0 & a_y - g_y & g_x - a_x & 0 & -\omega_x \omega_y & \omega_x^2 - \omega_y^2 & \alpha_x - \omega_y \omega_z & \omega_x \omega_y & \alpha_y + \omega_x \omega_z & \alpha_z \end{pmatrix} \quad (5)$$

noise and the bias sensitivity of the estimation results, the correlation matrix Υ consisting of M different V matrices compiled at subsequent instants of time

$$\Upsilon = \tilde{V}^T \tilde{V} \quad (12)$$

$$\tilde{V} = [V_1^T V_2^T \dots V_M^T]^T \quad (13)$$

plays a decisive rule. The sensitivity of $S\varphi$ to errors in $\tilde{V}$ can be shown to increase with increasing condition number $\kappa(\Upsilon)$ if the noise affecting $\tilde{V}$ and the force/torque measurements remains unchanged [6], [7]. The condition number $\kappa(X)$ of a square matrix X is defined as the ratio between the largest and the smallest singular value applying the 2-norm, i.e. $\kappa(X) = \|X\| \|X^{-1}\|$ [8].

Various methods to obtain optimized trajectories for robot dynamic parameters estimation exist that can readily be applied to inertial load parameters estimation. One popular approach optimizes $\kappa(\Upsilon)$ on a restricted set of functions, i.e. sine/cosine functions representing joint angles [9], [10]. Following this approach, a trajectory for joint i is given by:

$$q_i(t) = \sum_{k=1}^N \rho_{i,k} \sin(2\pi k f t) + \sigma_{i,k} \cos(2\pi k f t) + q_{i,0} \quad (14)$$

The parameter $q_{i,0}$ denotes a constant offset angle; $\rho_{i,k}$ and $\sigma_{i,k}$ are the coefficients of the sine and the cosine part respectively. N determines the number of superposed harmonics. The base frequency f of the trajectory is identical for all joints to ensure periodicity. Note that trajectories according to eq. 14 are intrinsically jerk-limited since the derivative of a sinusoidal function is sinusoidal again.

Apart from the constraints established in [11], i.e. minimum and maximum joint angles, maximum joint velocities, and maximum joint accelerations, compliance to the maximum ratings of the force/torque sensor, the acceleration sensor, and the angular velocity sensor has to be checked during optimization. Furthermore, environmental constraints, e.g. self-collisions of the manipulator, have to be taken into account. In order to comply with the maximum sensor ratings, each sample point of a candidate trajectory is checked for violations of the maximum ratings regarding accelerations and angular velocities. Concerning the force/torque sensor, a maximum mass, object dimensions and inertia matrix are assumed to test whether the forces acting on the sensor during trajectory execution comply with its maximum ratings. To minimize the condition number of Υ , following steps are performed: First, a Monte Carlo search is carried out to find promising candidate trajectories for optimization. Then, the `fmincon` function provided by the

Matlab Optimization Toolbox [12] is applied to each of those trajectories to minimize $\kappa(\Upsilon)$. At each optimization iteration the compliance to the constraints stated above is checked.

IV. PREPROCESSING

Before the parameter vector $S\varphi$ is estimated, two basic preprocessing steps have to be performed:

- Derivation of joint angular velocities and joint angular accelerations from the joint angle measurements provided by resolvers/encoders.
- Calculation of the angular velocity $S\omega$, angular acceleration $S\alpha$, and linear acceleration vector $S\mathbf{a}$ w.r.t. the sensor frame S based on the latter signals.

Due to the noise present in the joint angle measurements $\bar{q}_i(t)$, Kalman filters are applied to obtain joint angular velocities and joint angular accelerations. The utilized model is given by eqns. 15 and 16.

$$\mathbf{x}_{i,k} = \begin{pmatrix} 1 & \Delta t & 0 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{pmatrix} \mathbf{x}_{i,k-1} + \begin{pmatrix} 0 \\ 0 \\ p_i \end{pmatrix} \quad (15)$$

$$\bar{q}_i(t) = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \mathbf{x}_{i,k} + m_i \quad (16)$$

The state vector

$$\mathbf{x}_{i,k} = [q_{i,k}, \dot{q}_{i,k}, \ddot{q}_{i,k}]^T \quad (17)$$

consists of the joint angle $q_{i,k}$, the joint angular velocity $\dot{q}_{i,k}$, and the joint angular acceleration $\ddot{q}_{i,k}$ of joint i at time-step k . m_i and p_i denote the measurement noise and the process noise respectively; Δt designates the sampling interval. Based on these Kalman filtered signals, the angular velocity $S\omega$, angular acceleration $S\alpha$, and linear acceleration vector $S\mathbf{a}$ of the sensor frame are calculated propagating the contributions of each joint through the kinematic chain from frame $j-1$ to frame j according to eqns. 18 - 20 until reaching the sensor frame [13]. $\mathbf{z}$ stands for $\mathbf{z} = [0, 0, 1]^T$ and ${}^j_{j-1}\mathbf{R}$ designates a rotation matrix relating the orientation of frame $j-1$ to that of frame j .

$${}^j\omega = {}^j_{j-1}\mathbf{R} {}^{j-1}\omega + \mathbf{z} \dot{q}_j \quad (18)$$

$${}^j\alpha = {}^j_{j-1}\mathbf{R} {}^{j-1}\alpha + \mathbf{z} \ddot{q}_j + {}^j_{j-1}\mathbf{R} {}^{j-1}\omega \times \dot{q}_j \mathbf{z} \quad (19)$$

$${}^j\mathbf{a} = {}^j_{j-1}\mathbf{R} ({}^{j-1}\mathbf{a} + {}^{j-1}\alpha \times {}^{j-1}\mathbf{l}_j + {}^{j-1}\omega \times ({}^{j-1}\omega \times {}^{j-1}\mathbf{l}_j)) \quad (20)$$

where

$${}^{j-1}\mathbf{l}_j = \begin{pmatrix} a_j \cos(q_j) \\ a_j \sin(q_j) \\ d_j \end{pmatrix} \quad (21)$$

with Denavit-Hartenberg parameters a_j and d_j [13]. The gravity vector w.r.t. the sensor frame ${}^S\mathbf{g}$ is given by

$${}^S\mathbf{g} = {}^S\mathbf{R}_6 {}^6\mathbf{R}_5 {}^5\mathbf{R}_4 {}^4\mathbf{R}_3 {}^3\mathbf{R}_2 {}^2\mathbf{R}_1 (-g_0\mathbf{z}) \quad (22)$$

where $g_0 = 9.80665 \frac{m}{s^2}$.

V. PARAMETER ESTIMATION

Based on the equations presented in Section II the inertial parameters of the object may be estimated. To reduce estimation bias resulting from the noise characteristics of the employed sensors, the recursive instrumental variables (RIV) method is employed. In contrast to the recursive least-squares (RLS) method, the RIV method can yield bias-free estimates in the presence of correlated noise if the instrumental variables fulfill the requirements addressed in [14]. The equations of the employed variant of the RIV method are given by eqs. 23 - 25.

$$\mathbf{K}_k = \Sigma_{k-1} \mathbf{W}_k^T [\mathbf{U}_k \Sigma_{k-1} \mathbf{W}_k^T + \Lambda]^{-1} \quad (23)$$

$$\Sigma_k = [\mathbf{E} - \mathbf{K}_k \mathbf{U}_k] \Sigma_{k-1} \quad (24)$$

$${}^S\hat{\boldsymbol{\varphi}}_k = {}^S\hat{\boldsymbol{\varphi}}_{k-1} + \mathbf{K}_k \left[\begin{pmatrix} {}^S\mathbf{f}_k \\ {}^S\boldsymbol{\tau}_k \end{pmatrix} - \mathbf{U}_k {}^S\hat{\boldsymbol{\varphi}}_{k-1} \right] \quad (25)$$

Both $\mathbf{U}_k$ and $\mathbf{W}_k$ correspond to $\mathbf{V}$ but the signal sources of their elements may be different. The identification variables matrix $\mathbf{U}_k$ contains identification variables ${}^S\boldsymbol{\omega}_{id}$, ${}^S\boldsymbol{\alpha}_{id}$, ${}^S\mathbf{a}_{id}$ and ${}^S\mathbf{g}_{id}$ whereas the instrumental variables matrix $\mathbf{W}_k$ contains instrumental variables denoted by ${}^S\boldsymbol{\omega}_{inst}$, ${}^S\boldsymbol{\alpha}_{inst}$, ${}^S\mathbf{a}_{inst}$ and ${}^S\mathbf{g}_{inst}$. Λ is the measurement noise covariance matrix and Σ_k denotes the parameter covariance matrix.

Least-squares methods assume the additive error model in eq. 26:

$$\begin{bmatrix} {}^S\mathbf{f} \\ {}^S\boldsymbol{\tau} \end{bmatrix} + \mathbf{e} = \mathbf{V} {}^S\boldsymbol{\varphi} \quad (26)$$

Following Ljung [14], appropriate instrumental variables basically have to be correlated with their corresponding identification variables and uncorrelated with the error $\mathbf{e}$. Since the angular velocity, angular acceleration, and linear acceleration vectors of the sensor frame are provided by different signal sources, identification variables may be selected from one source (e.g. ${}^S\boldsymbol{\omega}_{id}$ provided the angular velocity sensor) and the corresponding instrumental variables from the other (e.g. ${}^S\boldsymbol{\omega}_{inst}$ originating from the joint angle measurements). Thus high correlation between the instrumental and the identification variables is assured.

Apart from sensor noise, the mounting plate of typical force/torque sensors is an error source. A simplified sketch of the arrangement of the sensing elements of common force/torque sensors is presented in Fig. 2. Clearly, the mounting plate (also referred to as distal sensor part) induces forces/torques into the sensing elements as any other object mounted *onto* the plate. Therefore, the inertial parameters of the distal sensor part have to be estimated in order to compensate these disturbances. Furthermore, the forces and torques caused by the gripper have to be considered. Since the inertia matrix of a complex body can be decomposed into the inertia matrices of its parts, the inertia matrices of

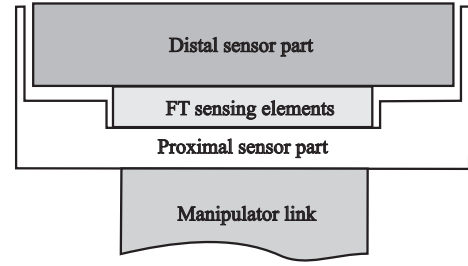


Fig. 2. Simplified structure of a common force/torque sensor.

the gripper $\mathbf{I}_{gripper}$ and the distal sensor part $\mathbf{I}_{dist}$ can be subtracted from the estimation result $\hat{\mathbf{I}}_{all}$ thus yielding the inertia matrix of the object.

$${}^S\hat{\mathbf{I}}_{object} = {}^S\hat{\mathbf{I}}_{all} - {}^S\mathbf{I}_{dist} - {}^S\mathbf{I}_{gripper} \quad (27)$$

Analogously, the remaining elements of the parameter vectors of the distal sensor part and the gripper can be subtracted from the estimated parameter vector ${}^S\hat{\boldsymbol{\varphi}}_{all}$.

$${}^S\hat{\boldsymbol{\varphi}}_{object} = {}^S\hat{\boldsymbol{\varphi}}_{all} - {}^S\boldsymbol{\varphi}_{sens} - {}^S\boldsymbol{\varphi}_{gripper} \quad (28)$$

VI. FEATURE EXTRACTION

Since features that are invariant w.r.t. rotation and translation are desirable, the mass and the principal moments of inertia are extracted from the complete set of inertial parameters. The parameter covariance matrix Σ provided by the RIV method can be beneficially utilized in the recognition process. Therefore, the covariance is propagated through the nonlinear feature extraction function using the scaled unscented transform (UT) [4] with the sigma point selection algorithm proposed in [15]. To obtain the desired features, the inertia matrix is first expressed w.r.t the estimated center of mass of the object employing the parallel-axis theorem, eq. 29, where $\mathbf{E}$ denotes the identity matrix again.

$${}^C\mathbf{I}_{object} = {}^S\mathbf{I}_{object} - m [({}^S\mathbf{c}^T {}^S\mathbf{c}) \mathbf{E}_{3 \times 3} - ({}^S\mathbf{c} {}^S\mathbf{c}^T)] \quad (29)$$

Secondly, estimates of the principal moments of inertia I_a, I_b , and I_c are determined by calculating the eigenvalues of the inertia matrix ${}^C\mathbf{I}_{object}$ assuring numerical stability with a balancing step. The resulting feature vector

$$\mathbf{f} = [\hat{m}, \hat{I}_a, \hat{I}_b, \hat{I}_c]^T \quad (30)$$

and its covariance matrix Σ_f obtained by the UT are used for object recognition.

VII. OBJECT RECOGNITION AND POSE ESTIMATION

The object recognition approach assumes that the features of each object of a set of objects are determined *once* in an arbitrary gripping pose. After this training phase, an object O can be recognized by determining the trained object Q whose feature vector $\mathbf{f}_Q$ has the smallest distance to the feature vector $\mathbf{f}_O$. The symmetric Kullback-Leibler divergence J_{KL} between two Gaussian densities [16] serves

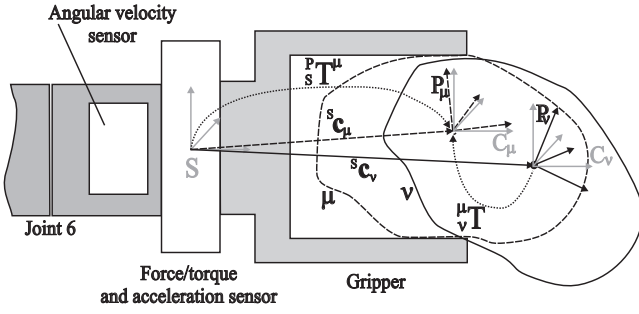


Fig. 3. Gripper with objects in two gripping poses.

as distance measure. It utilizes both the feature vector f_O and its covariance matrix Σ_{f_O} :

$$J_{KL}(O, Q) = \frac{1}{2} \left[(f_O - f_Q)^T (\Sigma_{f_Q}^{-1} + \Sigma_{f_O}^{-1}) (f_O - f_Q) + \text{tr} (\Sigma_{f_Q}^{-1} \Sigma_{f_O} + \Sigma_{f_O}^{-1} \Sigma_{f_Q} - 2E) \right] \quad (31)$$

Fig. 3 shows two gripping poses of an object related by a transform ${}^\mu_\nu T$. The frames P_μ and P_ν align with the principal axes of inertia in the respective object pose. To estimate a transformation ${}^S T^\mu$ from the sensor frame S to the frame P_μ , both a translation matrix ${}^S C^\mu$ shifting the sensor frame S to the COM C_μ of the gripped object in pose μ and a rotation matrix ${}^P C R^\mu$ aligning the resulting frame with the principal axes of inertia in pose μ have to be determined. The desired translation matrix ${}^S T^\mu$ is given by ${}^S T^\mu = \text{Trans}({}^S C_\mu)$. In order to determine the rotation matrix ${}^P C R^\mu$, the eigenvectors of the inertia matrix ${}^C \hat{I}_{object}^\mu$ are calculated. A right-hand basis B^μ constituted by the eigenvectors yields the desired rotation matrix. Note, however, that the right-hand basis given by the eigenvectors is not unique but four different right-hand bases exist, since changing the sign of two basis vectors yields a valid right-hand basis again. These alternative bases are related to each other by rotations of π around the basis axes. Clearly, these ambiguities do not allow for a unique pose estimate but yield alternative hypotheses. To obtain the transformation matrix ${}^\mu_\nu T$ relating the object pose ν to the object pose μ , equation 32 can be used,

$${}^\mu_\nu T = B_\nu^{-1} \text{Trans}({}^S C_\mu - {}^S C_\nu) B_\mu \quad (32)$$

where B_ν denotes a right-hand basis composed of the eigenvectors of ${}^C \hat{I}_{object}^\nu$. This transformation may be employed to transform frame P_ν to P_μ and may thus be used to correct gripping errors in assembly tasks prior to mating.

VIII. EXPERIMENTAL RESULTS

This Section presents the experimental results of the trajectory design, the pose estimation, and the object recognition approach. The employed setup consists of a Stäubli RX60 industrial manipulator with an open control architecture, a combined 6D force/torque and 6D acceleration sensor manufactured by JR3, and an angular velocity sensor employing three Analog Devices ADXR300 angular rate transducers. Fig. 4 shows the robot hand with the sensors as well as

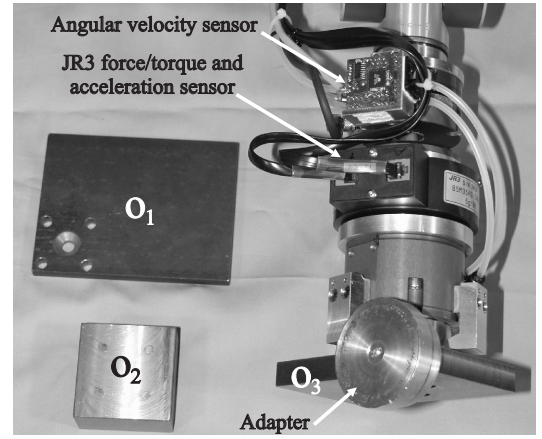


Fig. 4. Details of the experimental setup: Robot hand and test-objects.

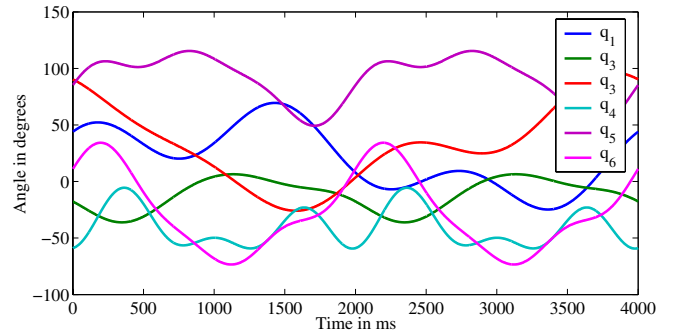


Fig. 5. Sinusoidal joint angles of trajectory T_2 .

three test-objects of *identical masses* but *different shapes* and a gripping adapter.

Firstly, major results of the trajectory optimization process are briefly discussed. Secondly, the parameter estimation results for the distal sensor part are presented. Thirdly, the performance of the inertial parameters estimation approach is evaluated. Then, the quality of the pose estimates is presented. Finally, the viability of the object recognition approach is assessed. Fig. 5 shows the joint angles of an optimized trajectory T_2 obtained with the optimization approach outlined in Section III. N has been set to 3 to confine the number of optimization variables. As can be seen, the trajectory is sinusoidal and periodic. Table I presents the simulated and the measured condition numbers associated with three optimized trajectories T_1 , T_2 , and T_3 . Clearly, the condition numbers originating from simulated identification variables (based on the *desired* joint angles of the trajectories) are considerably smaller than those based on experimental data. Moreover, the trajectory with the highest simulated condition number shows the smallest experimental condition number. The differences between the experimentally determined and the simulated condition numbers may be mainly attributed to the considerable differences [17] between the simulated and the measured acceleration signals. The dependence of the condition number on the test-object, however, may be neglected. Since the elements of the inertia matrix of the

TABLE I

SIMULATED AND EXPERIMENTALLY DETERMINED CONDITION NUMBERS
OF OPTIMIZED TRAJECTORIES WITH $N = 3$.

Trajectory	Test-object	Simulated κ	Experimental κ
T_1	o_1	7.72	23.23
T_1	o_2	7.72	23.42
T_2	o_1	8.18	16.89
T_2	o_2	8.18	14.44
T_3	o_1	6.51	18.64
T_3	o_2	6.51	18.85

TABLE II

ESTIMATED INERTIAL PARAMETERS OF THE DISTAL SENSOR PART OF
THE JR3 SENSOR 85M35A3-I40-D 200N12.

	m	c_x	c_y	c_z
Mean	56.5g	-1.7mm	0.4mm	10.3mm
Variance	0.017g ²	0.0067mm ²	0.0079mm ²	0.0098mm ²

distal sensor part are too small to be estimated accurately, static estimation is performed, i.e. merely the mass and the coordinates of the center of mass of the distal sensor part are determined employing different orientations of the sensor. Parameter estimation is carried out using the TLS method [8]. The parameter estimation results are presented in Table II. Due to the complex geometry of the gripper, its parameter vector $\hat{\varphi}_{gripper}$ is determined experimentally using the RIV method. The inertial parameters of the geometrically simple test-objects have been determined theoretically [18] from the mass of the objects and their dimensions thus establishing ground truth data. Note that for recognition each object carries two additional adapters whose inertial parameters have also been determined theoretically.

All following parameter estimation results are based on 1500 W and U matrices sampled at a rate of 1kS/s. Depending on the selected instrumental variables the estimation performance considerably differs. The results for the combinations of identification variables and instrumental variables compiled in Table III are presented in the following. Variables with index KF denote signals which are derived employing the Kalman filtered joint angle, joint angular velocity, and joint angular acceleration signals; signals originating from the force/torque, acceleration, or angular velocity sensors are denoted by index $sens$. Tables IV and V show the relative error of the estimates

$$e_{rel}(\hat{x}) = \left\| \frac{\hat{x} - x_{th}}{x_{th}} \right\| \cdot 100\% \quad (33)$$

where $\hat{x}$ denotes the estimated and x_{th} designates the the-

TABLE III

SELECTED IDENTIFICATION AND INSTRUMENTAL VARIABLE
COMBINATIONS FOR PARAMETER ESTIMATION.

Set	a_{id}	α_{id}	ω_{id}	a_{inst}	α_{inst}	ω_{inst}
s_1	a_{sens}	α_{sens}	ω_{sens}	a_{sens}	α_{sens}	ω_{sens}
s_2	a_{sens}	α_{KF}	ω_{sens}	a_{sens}	α_{KF}	ω_{sens}
s_3	a_{KF}	α_{KF}	ω_{sens}	a_{KF}	α_{KF}	ω_{sens}
s_4	a_{sens}	α_{KF}	ω_{sens}	a_{KF}	α_{KF}	ω_{KF}

TABLE IV

RELATIVE ERRORS OF THE MASS AND COORDINATES OF THE COM.

Set	$e_{rel}(m)$	$e_{rel}(c_x)$	$e_{rel}(c_y)$	$e_{rel}(c_z)$
s_1	1.23%	1.08%	3.21%	6.26%
s_2	1.61%	1.24%	3.49%	2.35%
s_3	0.70%	3.17%	6.35%	12.08%
s_4	0.42%	0.85%	1.05%	0.97%
$s_{4,n}$	0.61%	0.32%	1.71%	1.68%

TABLE V

RELATIVE ERRORS OF THE ELEMENTS OF THE INERTIA MATRIX.

Set	$e_{rel}(I_{xx})$	$e_{rel}(I_{xy})$	$e_{rel}(I_{xz})$	$e_{rel}(I_{yy})$	$e_{rel}(I_{yz})$	$e_{rel}(I_{zz})$
s_1	33.4%	15.57%	21.18%	29.24%	26.98%	11.51%
s_2	6.33%	18.82%	18.85%	0.53%	24.44%	0.81%
s_3	9.05%	13.11%	12.51%	9.89%	69.8%	12.32%
s_4	2.71%	5.73%	5.20%	6.14%	28.57%	2.22%
$s_{4,n}$	0.91%	9.73%	6.25%	6.42%	31.34%	1.62%

oretically determined value. The presented results originate from test-object o_3 (cf. Fig. 4) employing the trajectory T_2 (cf. Fig. 5) and a mounting pose P_1 . $s_{4,n}$ refers to the results of the parameter estimation without direct estimation of the offsets but zeroing in known pose according to Section II. The theoretically determined parameter vector of o_3 is:

$$\begin{aligned} \hat{\varphi}_{th} = & [1.00kg, -43.5kgmm, 28.5kgmm, 22.5kgmm, \\ & 2.21 \cdot 10^{-3}kgm^2, 1.22 \cdot 10^{-3}kgm^2, 0.96 \cdot 10^{-3}kgm^2, \\ & 3.74 \cdot 10^{-3}kgm^2, -0.63 \cdot 10^{-3}kgm^2, 4.95 \cdot 10^{-3}kgm^2]^T \end{aligned}$$

Sets s_1 - s_3 correspond to the RLS method since the instrumental variables equal the identification variables. Replacing the angular acceleration signals in s_1 by the signals based on the Kalman filtered joint angles (s_2) noticeably improves the estimation performance. However, using the linear acceleration signals based on the Kalman filtered joint angles as both identification and instrumental variables (s_3) deteriorates the estimation results. Estimation performance is clearly best if set s_4 is used. Regarding the approaches to offset handling, direct estimation of the offsets yields slightly better results. Calculating the principal moments of inertia for o_3 in three poses employing set s_4 yields the relative errors presented in Table VI. The evaluation of the gripping pose estimation is performed using one training pose P_1 and two estimation poses P_2 and P_3 . Table VII lists the true RPY angles relating the poses and the estimation results employing o_1 and o_3 . In order to assess the viability of the

TABLE VI

MASS, PRINCIPAL MOMENTS OF INERTIA AND RELATIVE ESTIMATION
ERRORS FOR o_3 .

	m in kg	I_a in kgm ²	I_b in kgm ²	I_c in kgm ²
$s_{I_{th}}$	1.000	$2.24 \cdot 10^{-3}$	$1.41 \cdot 10^{-3}$	$0.84 \cdot 10^{-3}$
Pose	$e_{rel}(m)$	$e_{rel}(I_a)$	$e_{rel}(I_b)$	$e_{rel}(I_c)$
P_2	0.41%	2.67%	9.21%	13.09%
P_3	1.16%	0.73%	10.64%	4.76%
P_4	0.93%	3.19%	16.32%	28.52%

TABLE VII

CALCULATED AND ESTIMATED ROLL, PITCH, AND YAW ANGLES OF ORIENTATION TRANSFORMS BETWEEN POSES.

Poses	Φ_z	Φ_y	Φ_x
$P_1 \rightarrow P_2$	90°	0°	0°
$P_1 \rightarrow P_3$	90°	90°	0°
o_1			
$P_1 \rightarrow P_2$	81.03°	-3.06°	-6.14°
$P_1 \rightarrow P_3$	86.23°	81.71°	3.86°
o_3			
$P_1 \rightarrow P_2$	78.62°	-12.32°	8.27°
$P_1 \rightarrow P_3$	78.69°	82.38°	14.26°

TABLE VIII

KULLBACK-LEIBLER DIVERGENCES BETWEEN TEST-OBJECTS IN DIFFERENT POSES.

Pose	J_{KL,o_1}	J_{KL,o_2}	J_{KL,o_3}
o_1			
P_2	31.14	98.81	167.42
P_3	39.38	119.23	187.28
P_4	21.18	107.35	214.77
o_2			
P_2	108.87	32.65	217.49
P_3	94.58	15.54	198.10
P_4	112.33	49.31	247.46
o_3			
P_2	175.52	247.27	15.53
P_3	196.42	223.18	12.28
P_4	201.14	194.36	31.41

object recognition approach, the objects are equipped with two adapters to allow for various gripping poses. In a first step, the features of all three test-objects are determined in an arbitrary training pose. In the following recognition step, the symmetric KL-divergences between the examined object and the test-objects in their training poses are calculated. Table VIII presents the results. Each of the columns two to four holds the KL divergence J_{KL,o_i} of the examined object in the pose listed in the first column to the test-object i in its training pose. As can be seen, the respective object is recognized correctly regardless of the used gripping pose.

IX. SUMMARY AND CONCLUSIONS

An object recognition and pose estimation approach for rigid objects employing estimates of inertial parameters is presented. The employed excitation trajectories are optimized w.r.t. parameter excitation. However, the condition numbers based on simulated data are shown to disagree with the condition numbers based on experimental data. Therefore, the optimization of trajectories based on the *simulated* condition number is questionable if the actual trajectory of the manipulator deviates from the simulated one. However, such deviations virtually always occur due to unmodelled dynamics effects. The angular acceleration signals provided by the wrist-mounted acceleration sensor prove to be unsuitable for parameter estimation due to considerable noise and disturbances whereas the angular acceleration signals derived from encoder measurements lead to promising estimation

results. The estimation results of the inertial parameters and the principal moments of inertia show comparatively low relative errors [2]. Especially concerning the pose estimates, the accuracy of the approach may be increased by utilizing estimation approaches that do not only consider errors in the measured forces/torques but also in the identification variables (e.g. recursive variants of the TLS approach) [17]. The object recognition approach proves to robustly recognize the three test-objects despite having identical masses.

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